Innovation, entrepreneurship and business models 5:

The context for innovation, diffusion and adoption

Christian R. Østergaard

Today

- The lecture will examine the context for innovation, as well as the processes of **diffusion** and **adoption** of new ideas (chapter 6). It will also explore how **intellectual property rights** (IPR) influence these processes (chapter 13).

Literature:

- *Tim Mazzarol, Sophie Reboud (2020) Entrepreneurship and Innovation: Theory, Practice and Context, 4th ed. Series: Springer Texts in Business and Economics. Chapter 6 and 13*

Last lecture on innovation in small and large firms

- Innovation in small and large firms
  - Differences in opportunity recognition and resource accumulation
  - Large firm
    - Often a more strategic approach to innovation
    - Continuous improvements
    - Multiple projects
    - Highly educated employees, R&D spending, dedicated resources to innovation
    - Large networks
    - More bureaucracy, exploitation versus exploration dilemma
  - SME
    - Less strategic and more opportunity driven, Stronger involvement of management
    - Fewer resources
    - Rely of formal/informal networks to acquire needed resources
    - Flat organization and flexibility
    - Incremental and radical innovations
- Capacity building for innovation capture
  - Strategy, IPR protection, Sales
- Today : processes of **diffusion** and **adoption** of new ideas, and IPR protection

Coca-Cola is pulling its newest ‘permanent’ flavor from store shelves

By Jordan Valinsky, CNN
3 minute read · Updated 8:07 AM EDT, Tue September 24, 2024

- Coca-Cola is discontinuing its Spiced flavor just six months after its launch, due to disappointing sales.
- The company plans to introduce a new flavor in 2025. Spiced, which blended traditional Coke with raspberry notes, was aimed at Gen-Z drinkers but failed to gain traction.
- Coca-Cola intended Spiced to become a permanent addition to its lineup because its research found that an “increase in consumer willingness to try a spiced beverage” and that raspberry was selected more than 5 million times on its Freestyle drink machines in 2022, which the company “often uses for inspiration” for new flavors
- Coca-Cola has been experimenting with various limited-time flavors to attract younger consumers.
- Despite initial high hopes, Spiced will be phased out as part of Coca-Cola’s ongoing strategy to refresh its product lineup.
- Drinkers are shifting their preferences away from sugary sodas and to sparkling waters and hydration beverages. In response, the company is growing its Topo Chico water line and expanding its BodyArmor brand.

Innovation Dynamics: Small vs. Large Firms

- Small and large firms **allocate resources** differently for innovation endeavors, influenced by financial capabilities, access to talent, and risk tolerance.
- The **organizational structure and processes for innovation** can vary between small and large firms, impacting agility, decision-making, and adaptability.
- Small firms **may** employ more agile and informal methods in their **search for new ideas**, while large firms **may** utilize structured processes and dedicated research teams.
- **Collaboration patterns** differ, with small firms often engaging in close-knit partnerships and large firms leveraging extensive networks and strategic alliances.
- Both small and large firms can produce various **types of innovation outcomes**, including incremental improvements, disruptive breakthroughs, and novel business models.
- **Managers** in large firms may prioritize efficiency and risk management, while managers in small firms often embrace experimentation and agility to drive innovation.

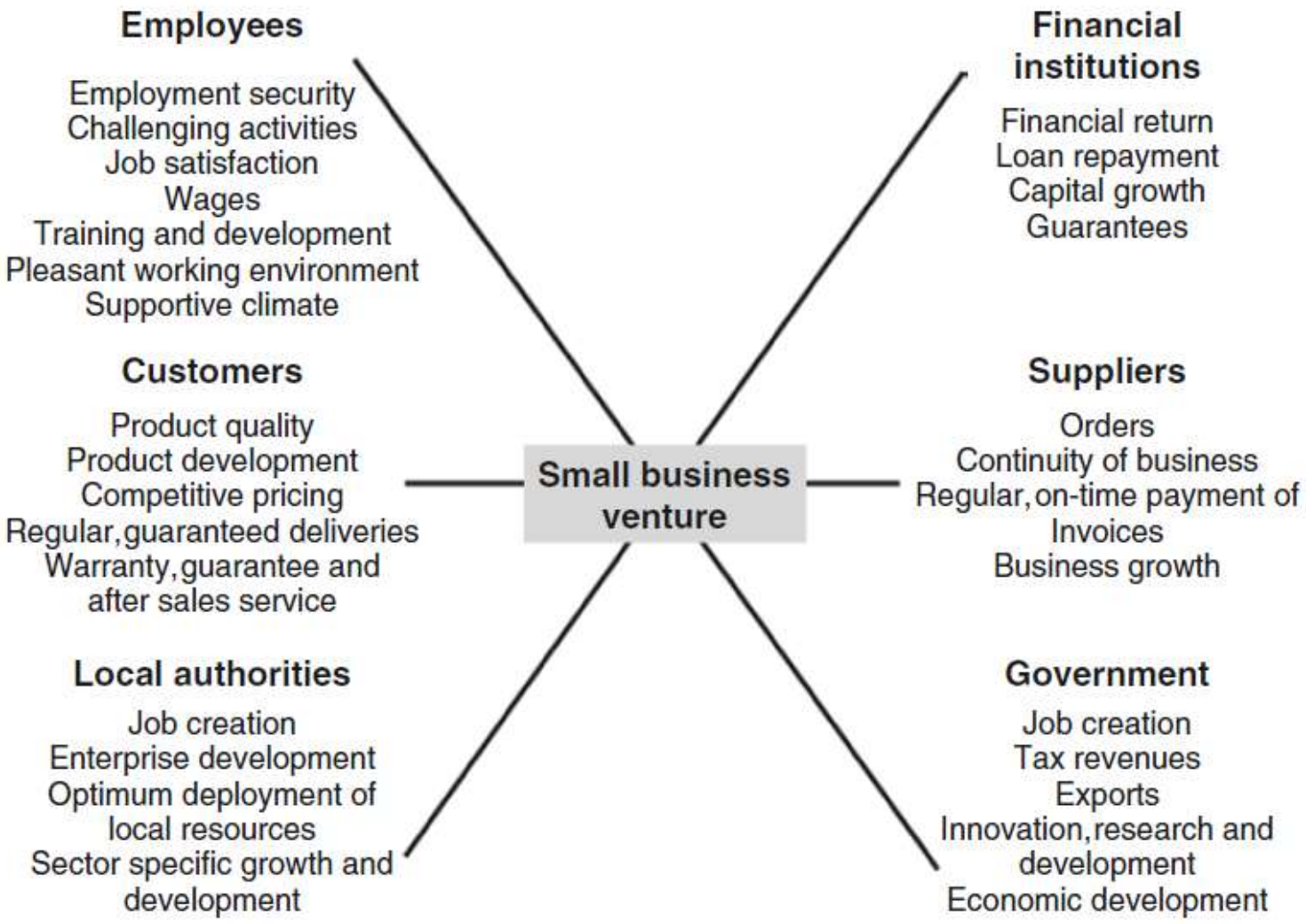


Fig. 5.5 Stakeholder web of the small firm. (Source: Jennings and Beaver 1997)

Strategic networking

- Potential sources of innovation
- They are more or less formal and more or less interactive
- Internal
- External
  - Customers, consumers and users
  - Suppliers
  - Competitors
  - Universities and technological institutes
  - Public-Private Partnerships

Properties of knowledge

- Need for knowledge
  - How to access the needed knowledge
  - Develop it yourself or collaborate
- Codification of knowledge
  - Tacit vs. codified knowledge
  - Tacit – hard to transfer - requires interaction
  - Codified knowledge – easier to transfer – requires prior knowledge/absorptive capacity
- Different types of knowledge (Johnson and Lundvall, 1994)
  - Know-what
  - Know-why
  - Know-how
  - Know-who

Absorptive Capacity

- absorptive capacity, involves the ability of the firm to leverage existing stocks of knowledge, to **acquire**, **absorb** and then **make effective use** of new knowledge. (Cohen and Levinthal, 1990)
- it must possess a culture and structure that encourages and facilitates its employees to possess the necessary mindsets and competencies to acquire, assimilate, transform and exploit knowledge, (Zahra and George 2002).
- This requires not only the firm's ability to effectively connect and communicate with **external** actors in the exchange of knowledge, but to also connect and communicate with **internal** actors in this knowledge exchange process (Cohen and Levinthal 1990).

Open innovation

- Open versus Closed Innovation Models:
- Closed Innovation: In the closed innovation model, organizations primarily rely on their internal research and development (R&D) capabilities to generate proprietary knowledge.
- Open Innovation: Contrary to closed innovation, the open innovation model involves expanding the organization's boundaries to facilitate the free exchange of ideas and knowledge. This approach acknowledges that businesses cannot afford to wait for internal technologies to develop. Instead, they actively seek out needed expertise and technologies, whether from their own research labs or external sources.
- According to Chesbrough (2003), companies should access necessary knowledge promptly, either from their internal research labs or from external sources.
- Key Issues in Open Innovation Management: An essential consideration in the **management of open innovation is the control of intellectual property (IP) rights**. Rather than viewing IP rights solely as protectors of a firm's patents, managers should adopt **a perspective of trading in IP rights for maximum value capture.**
- This may **involve licensing IP rights to other firms, including competitors, to leverage the firm's intangible assets effectively.**
- Additionally, successful implementation of open innovation strategies often requires dedicated staff training in IP rights management and the establishment of well-designed protocols to safeguard intangible assets and manage the dissemination of research findings.

Characteristics of innovative firms

- Search and sources of innovation
  - Innovative firms adopted practices to put them in the most competitive markets and in contact with the most demanding customers.
- Absorptive capacity and open innovation
  - innovative firms used their internal capacity to identify, gain access to and use the external environment for expertise and knowledge.
- Absorptive capacity and the innovative organisation
  - The skills of employees within innovative firms were valued. Interaction between skilled workers with technical skills enhanced creative opportunities and the generation of new ideas.

The Innovative Organisation

- **Market Orientation**: A focus on understanding and meeting customer needs through proactive market analysis and customer-centric decision-making.
- **Innovative Leadership**: Leadership that fosters a culture of creativity, experimentation, and risk-taking to drive continuous innovation throughout the organization.
- **Non-linear Strategic Planning**: Adopting flexible and adaptive approaches to strategy formulation and implementation to respond effectively to dynamic and uncertain business environments.
- **Ambidextrous Structure**: Balancing the exploitation of existing capabilities with the exploration of new opportunities through distinct but complementary organizational units or processes.
- **Innovation-Focused Culture**: An environment that values and encourages creativity, experimentation, and continuous improvement, fostering a collective mindset of innovation across the organization.

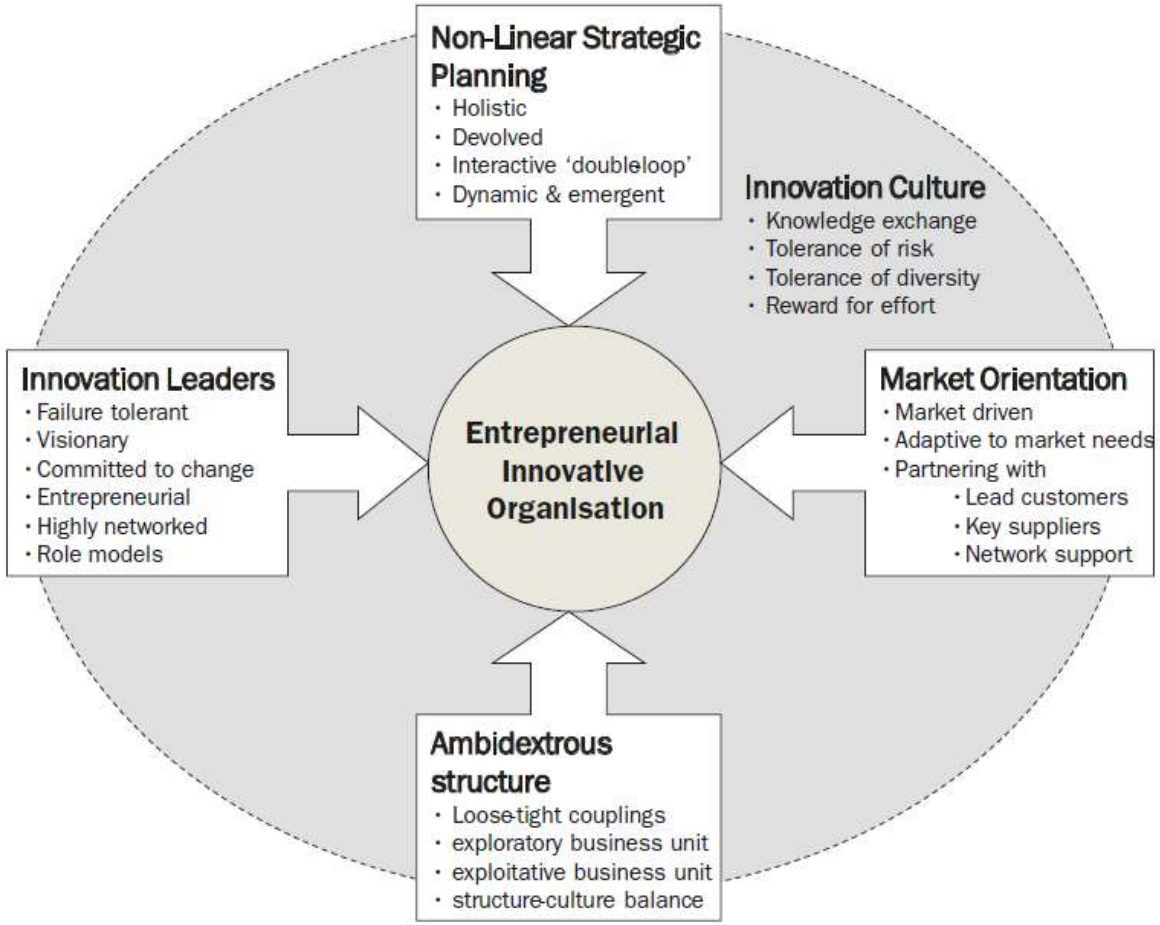
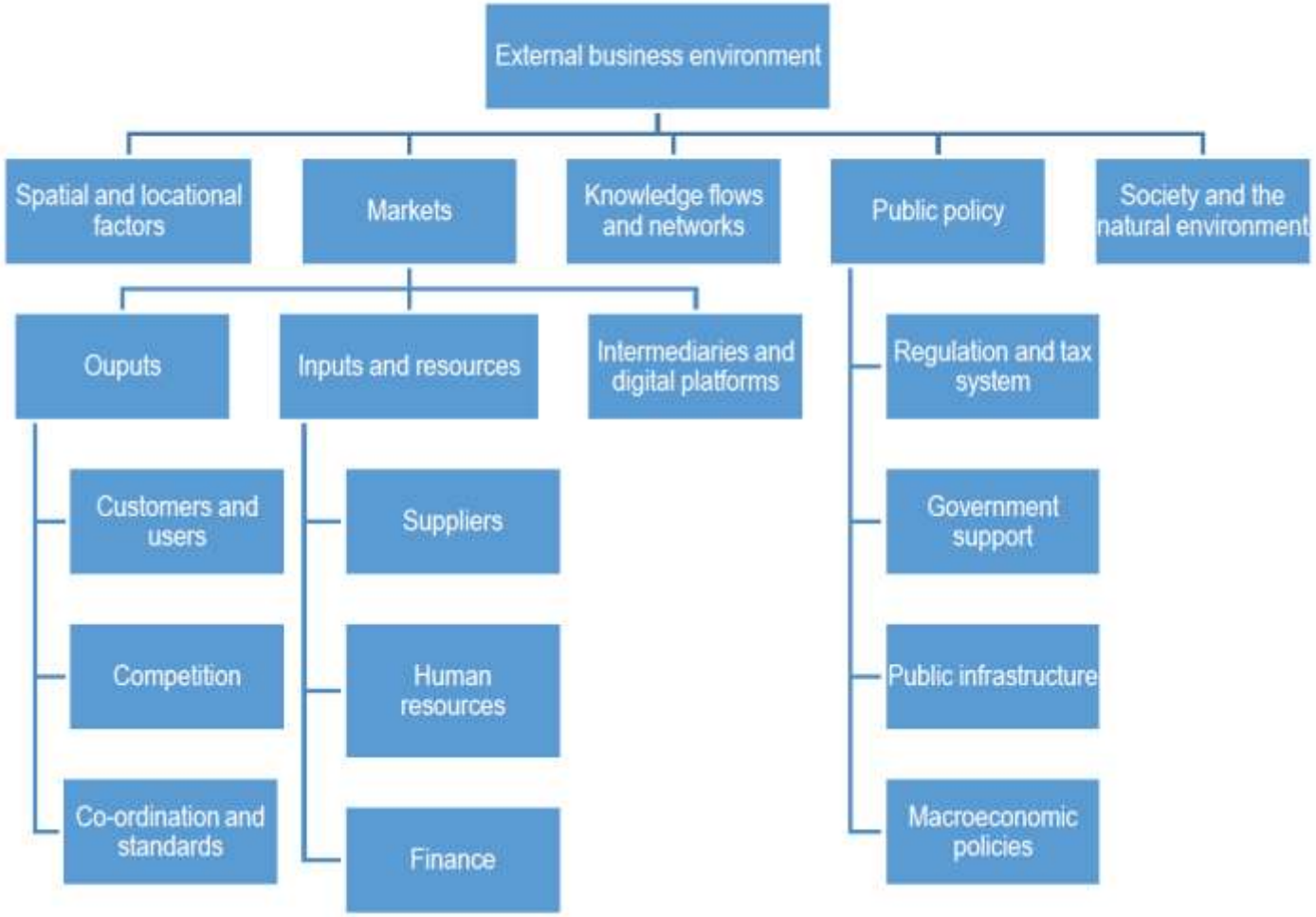


Fig. 4.4 A model of entrepreneurial innovative organisations. (Source: Mazzarol 2004)

The context for innovation – OECD (2018)

Figure 7.1. Main elements of the external environment for business innovation



What about diffusion and adoption and the role of IPR?

Adoption and diffusion of innovations

- Innovation diffusion is a social and economic process
  - importance of initial customer selection
  - pathways to market
  - barriers to market entry
  - substitution threats

Three innovation paradigms – **historical** archetypes

	The Creative Genius	*Technology Push*	*Market Pull*
Explanation of innovation	Psychological	Technological	Sociological
Determinant of innovation	Entrepreneurship	Technological development	Market-oriented Strategy
Agent of Innovation	The Gründer ('Amateur')	Scientist/ Engineer	The professional Manager
Result	Economic Growth & Corporate Development		

Fig. 6.1 Three innovation paradigms. (Source: Sundbo 1998)

1. The **creative genius** is motivated by psychological factors to pursue a passion or dream stemming from their own creative imagination, or in response to a problem or market opportunity.

2. The **scientist**: technology-push paradigm is characterized by a predetermined plan to follow established research methodologies and formal patterns of new technology development, less influenced by entrepreneurship.

3. The **professional manager** prioritizes satisfying customer or market demand and achieving relatively fast returns on investment in any innovation. They are more concerned with the ease of market acceptance than the technological brilliance of the innovation.

Different phases of an innovation process

- Internal, external or interactive processes
- Idea generation could also be a product of external collaboration-
  - Marketing forces derived from the customer or from attempts to solve customer problems are often the most significant source of commercially-useful ideas.
  - Role of creativity
- Marketing and commercialization
- -diffusion and adoptation

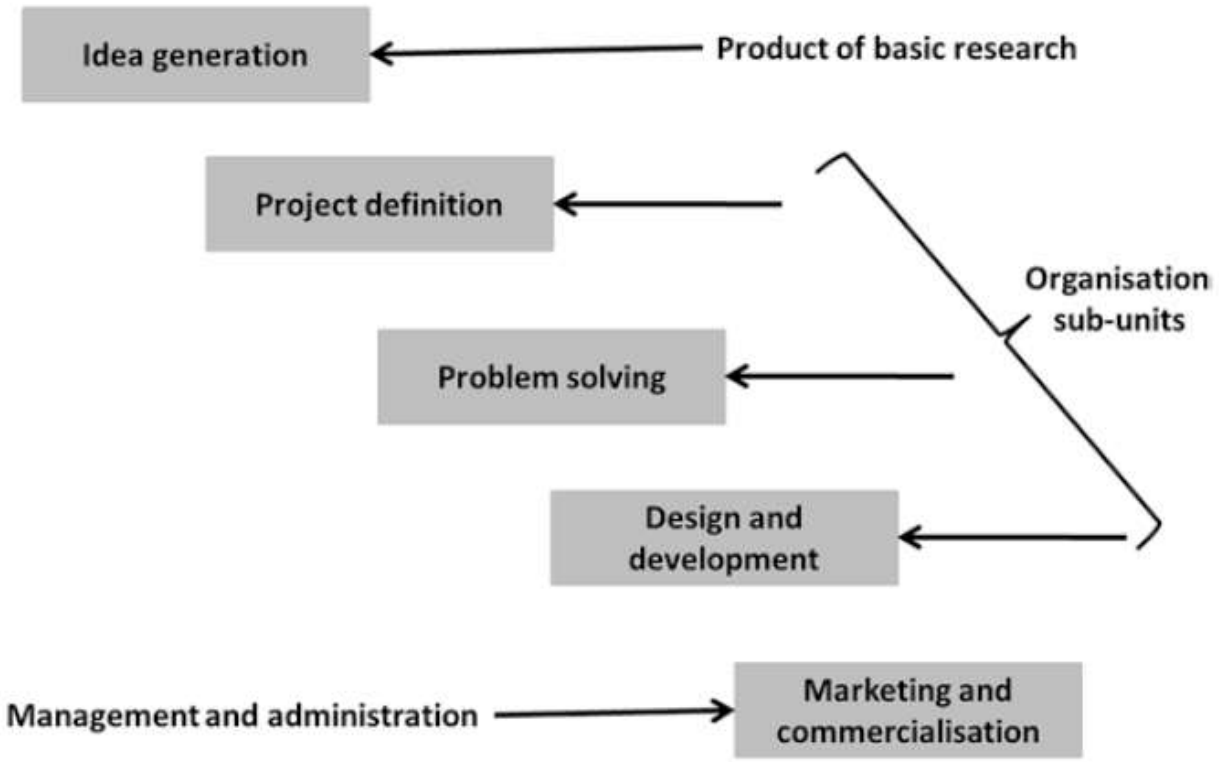


Fig. 6.2 Generating innovation

Diffusion processes – impact on society

- Diffusion refers to the process by which an innovation is communicated through different channels over time among members of a social system.
- The effect of technological change on economic growth depends upon the degree of innovation diffusion.
- Key issues (Rogers)
  - Does the innovation offer any significant relative **advantage** over existing technologies or processes?
  - Is it of a complexity level that makes it **difficult to use** or employ?
  - Is the new innovation **compatible** with existing technologies or processes, or will it force these to be **replaced**?
  - Can the new innovation be subject to **trial before adoption** and is it something that can be observed in practice before adoption?

Technology S-Curves

S-Curves in Technology Diffusion.

- Adoption is initially slow because the technology is unfamiliar.
- It accelerates as technology becomes better understood.
- Eventually market is saturated and rate of new adoptions declines.
- Technology diffusion tends to take far longer than information diffusion.
  - Technology may require acquiring complex knowledge or experience.
  - Technology may require complementary resources to make it valuable (for example, cameras not valuable without film or streaming without broadband)

The industry lifecycle revisited

- Example streaming
  - Moving from DVDs to streaming
- Experiments with pricing, subscription, ads/no ads, binge-dropping vs. weekly releases, licensed vs. original content
- Dominant design- the Netflix model- vertically integrated design (massive library and original series, monthly subscription)
- Now bundle deals, better algorithms, etc.

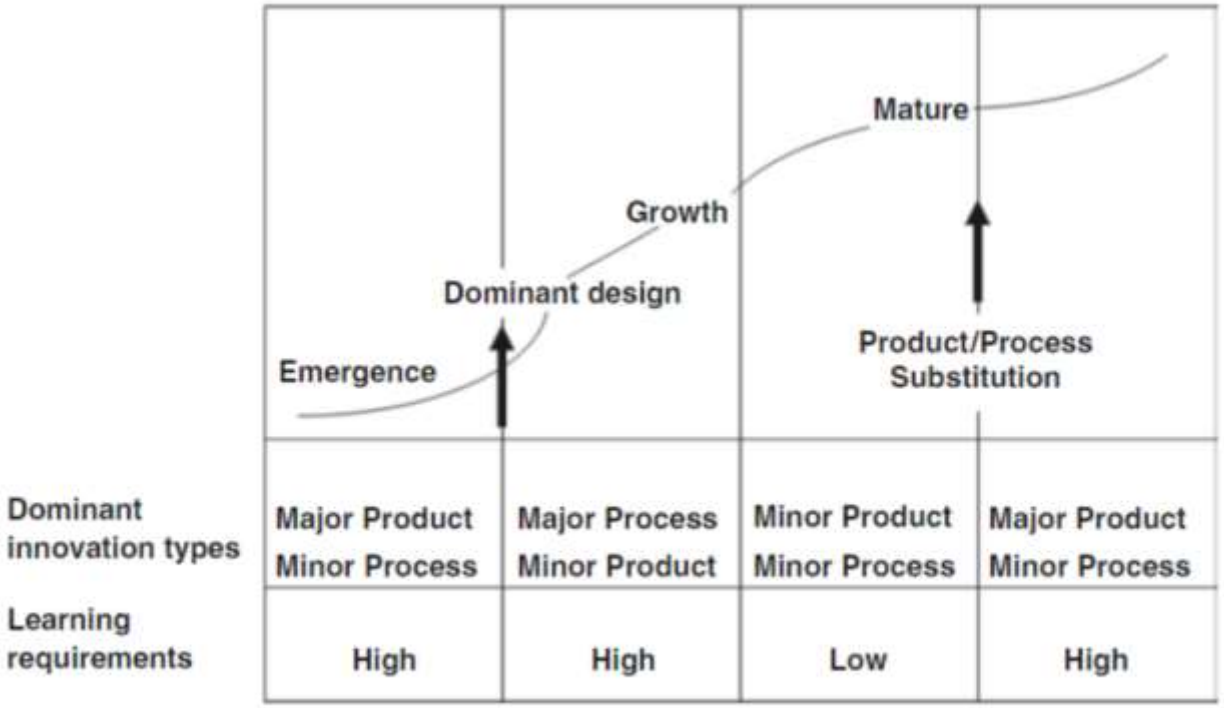


Fig. 1.10 Innovation lifecycles. (Source: Tushman and Nadler 1986)

Adoptation and diffusion is not easy or straightforward

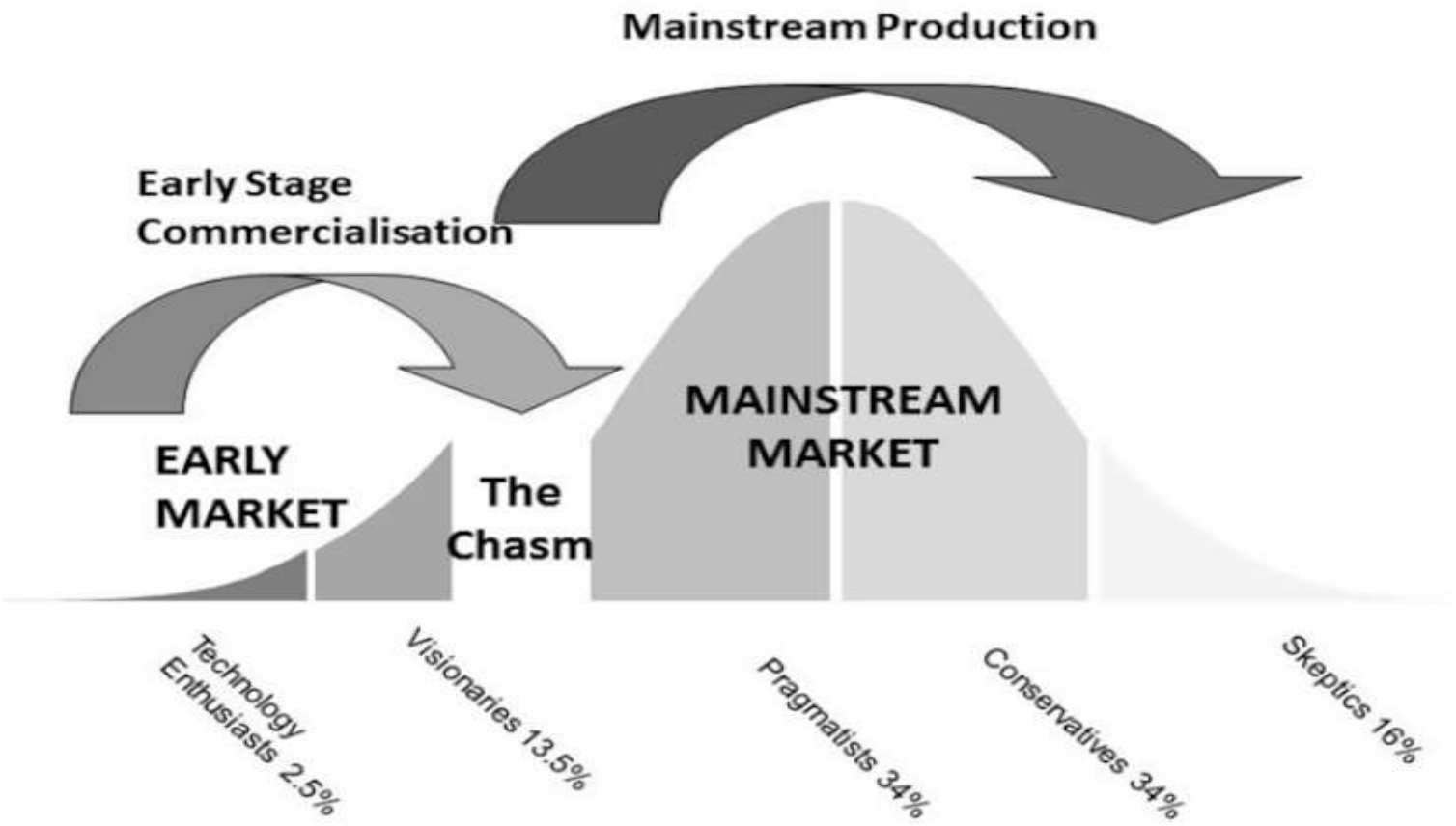


Fig. 6.3 Innovation diffusion model. (Source: Moore 2014)

Phases of adoption

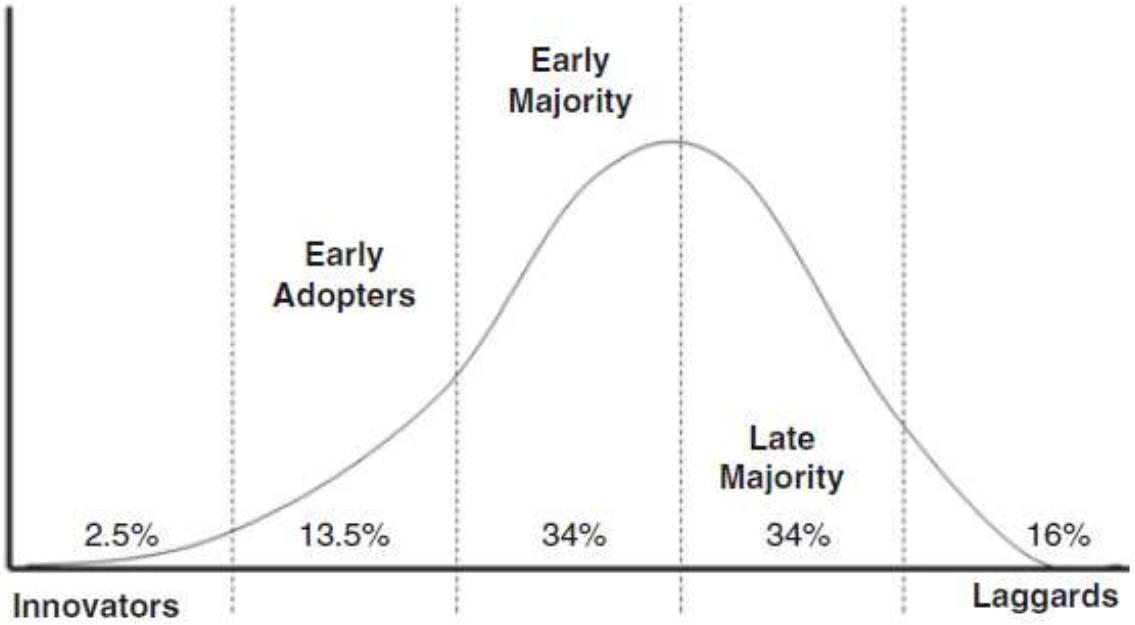
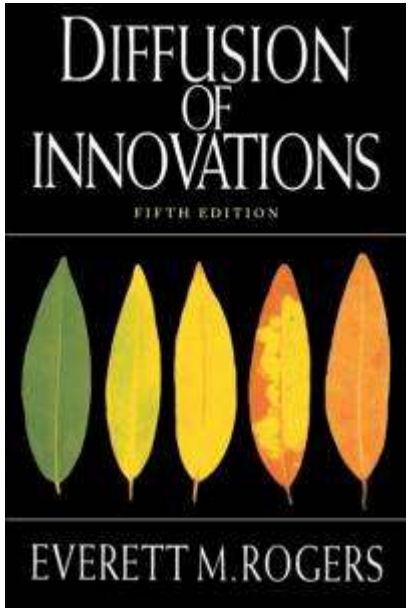


Fig. 6.9 Innovation diffusion-adoption patterns. (Source: Rogers 1962, 1995)



Exercise on smart phones

- The smart phone market and its industry life cycle.
  - Which phase is it in and why?
- What is a dominant design
  - Do we observe a standardized blueprint or set of features that most smart phones adhere to?
  - If so, how might this influence competition and innovation within the market?
- What characterise the industry today

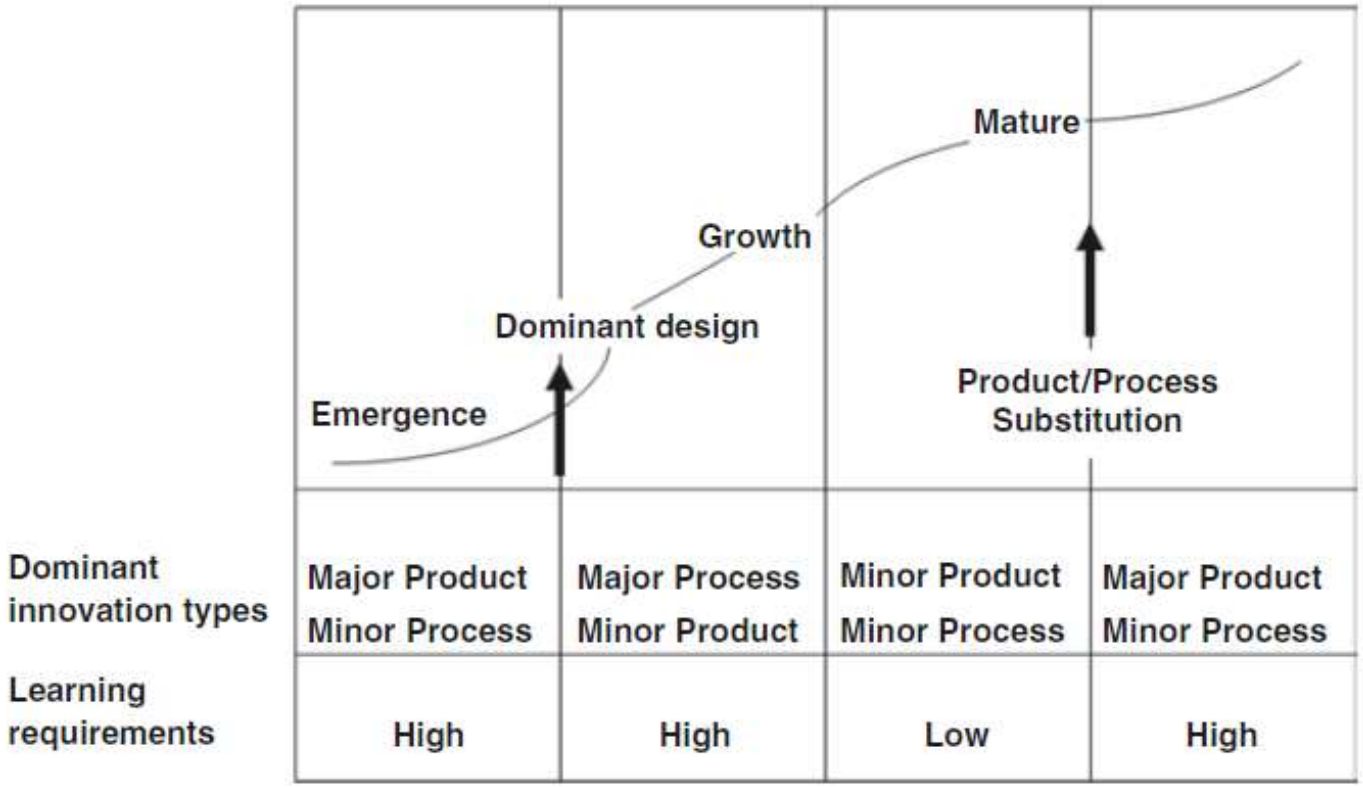


Fig. 1.10 Innovation lifecycles. (Source: Tushman and Nadler 1986)

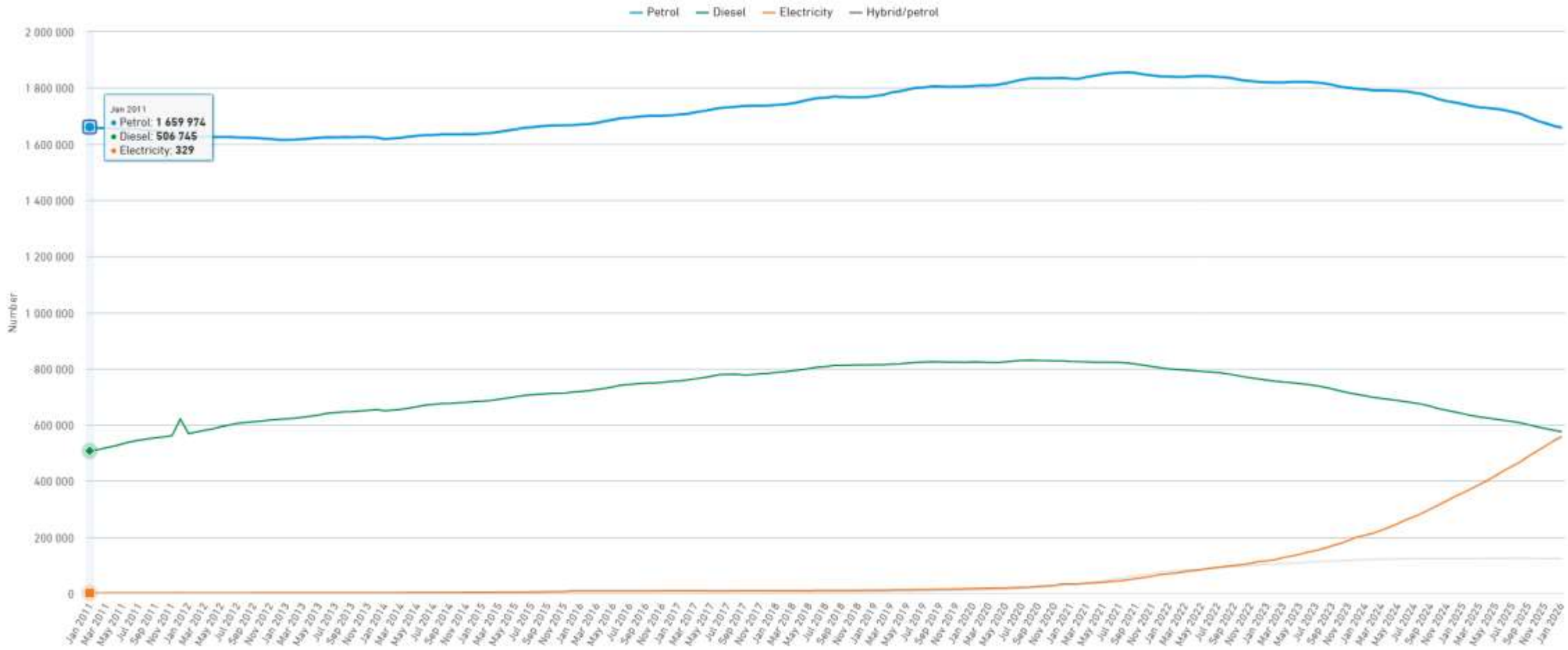
Diffusion of EVs

- Significant relative **advantage** over existing technologies or processes?
- Complexity level that makes it **difficult to use**?
- **Compatible** with existing technologies or processes, or will it force these to be **replaced**?
- **Trial before adoption** and is it something that can be observed in practice before adoption?
- What can be done to in order to speed up diffusion?
- Who can do something to speed up the process?
    - Individuals, Government,Electric car manufactures

- https://www.statista.com/topics/3213/electric-mobility-in-europe/

Stock of passenger cars

Ownership: Total | Propellant:



A central theme in innovation studies is the difference between how small and large firms innovate. Based on the course literature, which of the following statements accurately compare their typical innovation characteristics?

Choose one or more answers

- [ ] Small firms are generally characterized by rigid organizational structures and formal innovation processes to maximize efficiency.
- [ ] Large firms often rely on their flexibility, agility, and informal networks to seize opportunities, often in niche markets.
- [ ] Large firms typically possess greater financial resources and formal R&D departments, allowing them to pursue large-scale, capital-intensive innovation projects.
- [ ] The innovation process in a small firm is often heavily influenced by the personal characteristics and vision of the founder/entrepreneur.
- [ ] Large firms, due to their established routines, are inherently better at producing disruptive innovations than small firms.

Adoption models – Technology acceptance model

- Diffusion is contingent upon the **attitudes the adopters** hold toward the innovation, and these in turn influence their behavioral intention to adopt or 'dis-adopt'.
- Influencing the adopter's attitudes are their perceptions of the **ease** with which the innovation will be **used** and also how **useful** the new technology is likely to be.

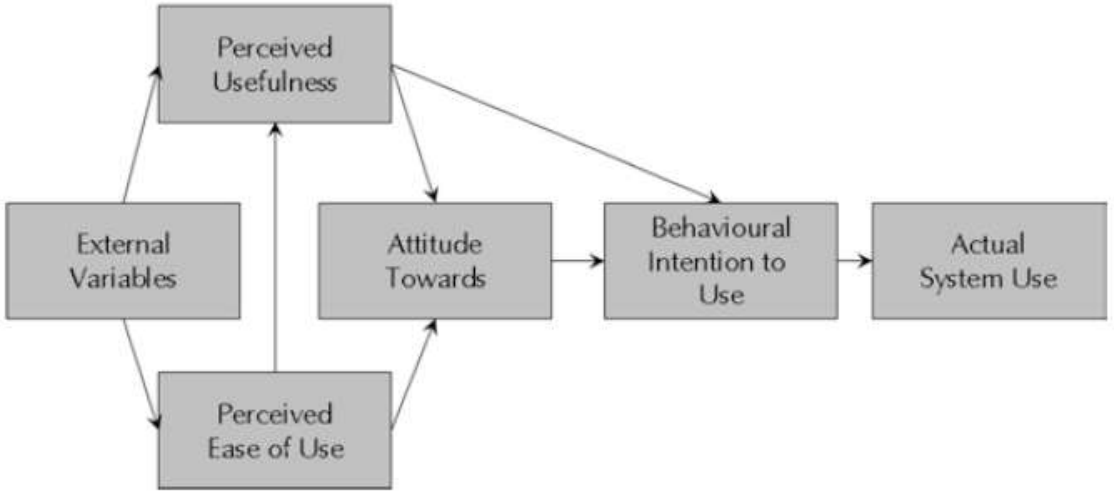


Fig. 6.5 Technology acceptance model (TAM). (Source: Davis et al. 1989)

TAM 2

- the influences of third parties
- how the innovation is viewed in terms of its usefulness
- how strongly the innovation impacts on the adopter's own work environment
- whether the benefits of the innovation can be quantifiably measured
- whether the results can be shown or demonstrated prior to adoption

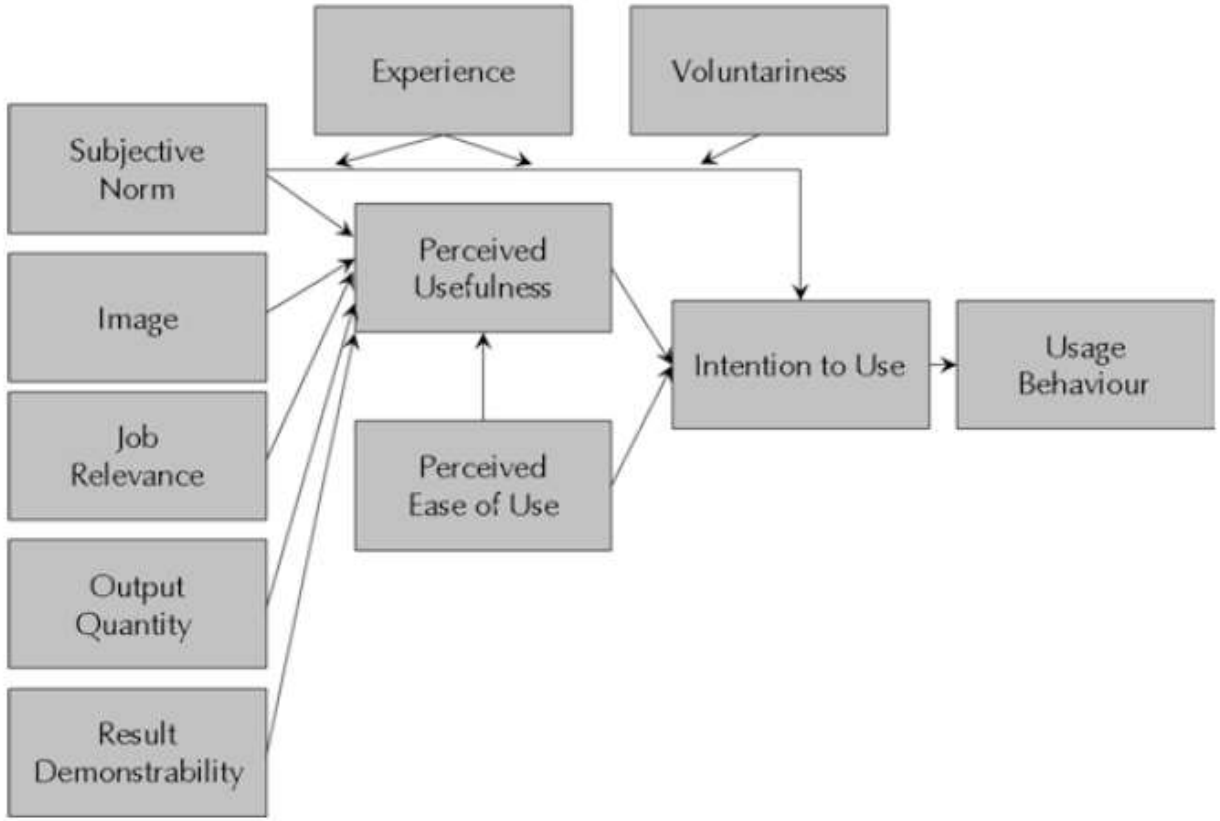


Fig. 6.6 Technology acceptance model 2 (TAM2). (Source: Legris et al. 2003)

Innovation Adoption **in** Organisations

- Two distinct stages
  - initial decision by senior management to adopt the innovation
    - Imposed or willingly accepted
    - sufficient support to assist the end-users
    - provision of training and support to the end-users
  - how the innovation is then adopted and assimilated into the organization
    - Subjective norms – influence of peers, customers, networks
    - Organisational culture
    - level of education and knowledge within the workforce
- Assimilation
  - **Initiation** – the innovation is launched
  - **Adoption** – the innovation is put into trial
    - adaptation – all end-users learn to deal with it and adapt work practices or systems
  - **Acceptance** – the innovation is accepted after trial
    - routinization – the new technology is now mainstream
  - **Infusion** – following regular use, the innovation may spur new opportunities or uses.

Exercise :Implementation of AI in firms

- *New York CNN —*
- Corporate America is rapidly adopting artificial intelligence to automate work once exclusively done by humans.
- More than half (61%) of large US firms plan to use AI within the next year to automate tasks previously done by employees, according to a survey of finance chiefs released Thursday. Those tasks include everything from paying suppliers and doing invoices to financial reporting, said the survey conducted by Duke University and the Federal Reserve Banks of Atlanta and Richmond. That's in addition to creative tasks for which some businesses are already relying on ChatGPT and other AI chatbots to assist, including crafting job posts, writing press releases and building marketing campaigns. The findings show companies are increasingly turning to AI to cut costs, boost profits and make their workers more productive. "You can't be running an innovative company without seriously considering these technologies. You run the risk of being left behind," Duke finance professor John Graham, academic director of the survey, told CNN in a phone interview. The CFO Survey, a collaboration of Duke and the Atlanta and Richmond Fed banks, found that nearly one in three (32%) firms — large or small — plan to use AI in the next year to complete tasks once done by humans.
- **Why bosses are deploying AI**
- Some of this is already happening — especially among larger firms that have the financial firepower to experiment with AI. Nearly 60% of all companies (and 84% of large companies) surveyed said that over the past year they have already leaned on software, equipment or technology including AI to automate tasks employees previously did. The survey was conducted between May 13 and June 3. Bosses are turning to AI for a variety of reasons, including to trim what they are spending on human workers.
- The CFO Survey found that companies say they are using automation to increase product quality (58% of firms); increase output (49%), reduce labor costs (47%) and substitute for workers (33%). Still, the good news for workers is that some experts don't believe AI will cause mass job loss, at least not right away.
- "I don't think there will be a lot of job loss in the year," said Graham. "In the short run, this will be more about plugging some holes and possibly not hiring someone they would have otherwise — but not laying someone off. In part that's because this is all-brand new."
- https://edition.cnn.com/2024/06/20/business/ai-jobs-workers-replacing/index.html
- **Discuss** How could absorptive capacity relate to adoption of AI in firms?
- Use the TAM model to discuss: What do you think influence the decision to adopt AI in firms?

Diffusion of innovations in society

1. Relative advantage
2. Complexity
3. Comparability
4. Trial-ability
5. Observability
6. Usefulness and ease of use
7. Subjective influence

- **Potential network effects**
- earlier adopters affect later adopters in the adoption decision
- later adopters affect earlier adopters in the dis-adoption of an innovation.
- S-curved diffusion models

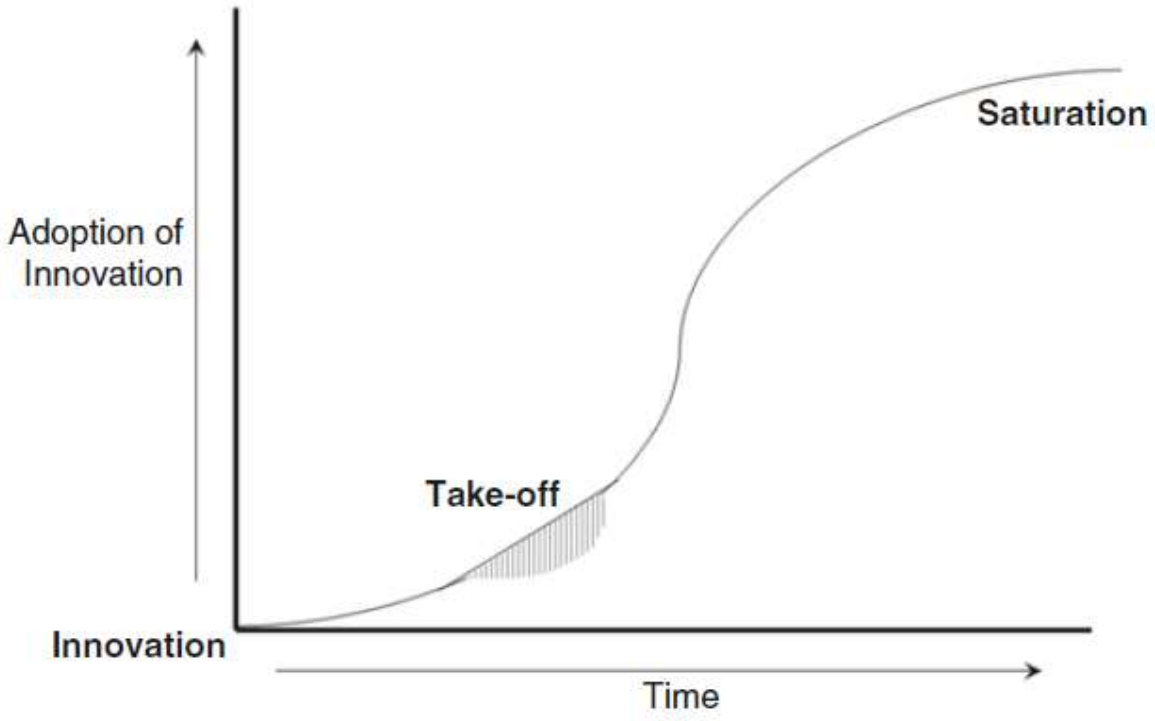


Fig. 6.7 Innovation diffusion model (sigmoid curve model)

Remember the phases of adoption

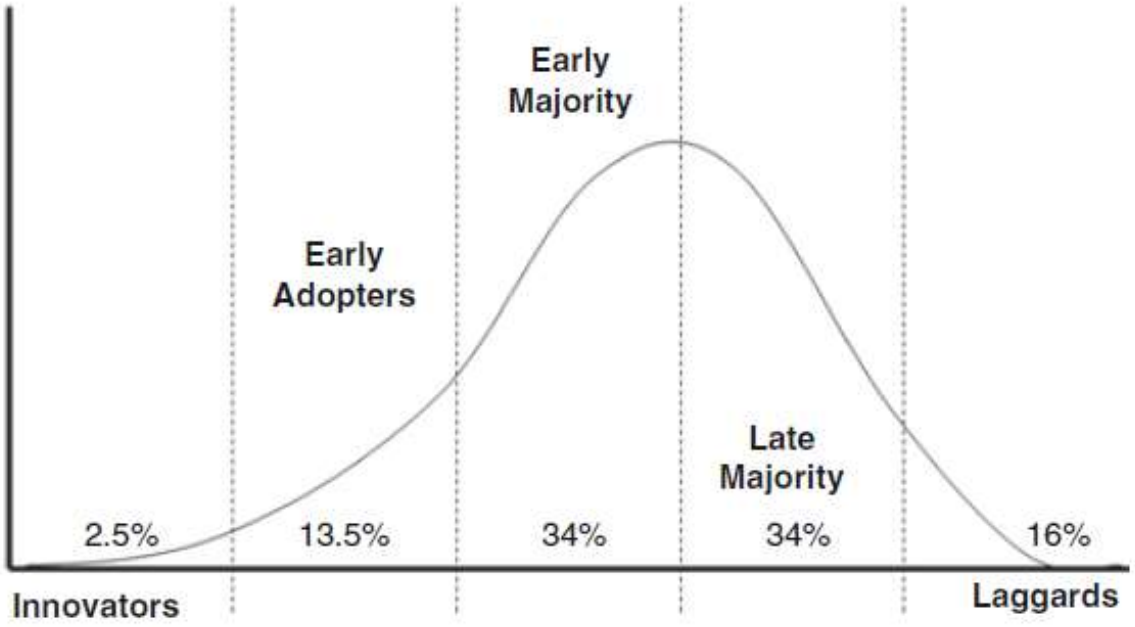
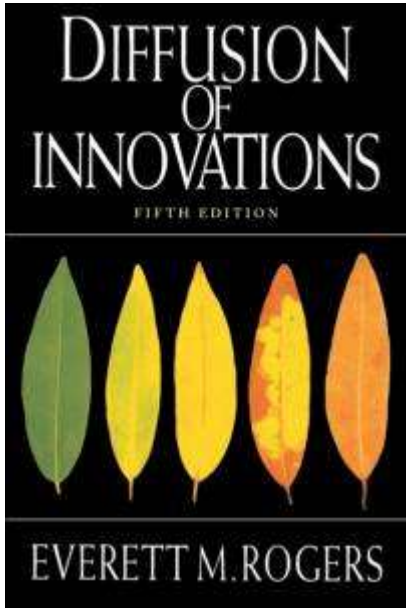


Fig. 6.9 Innovation diffusion-adoption patterns. (Source: Rogers 1962, 1995)



S-curved diffusion models

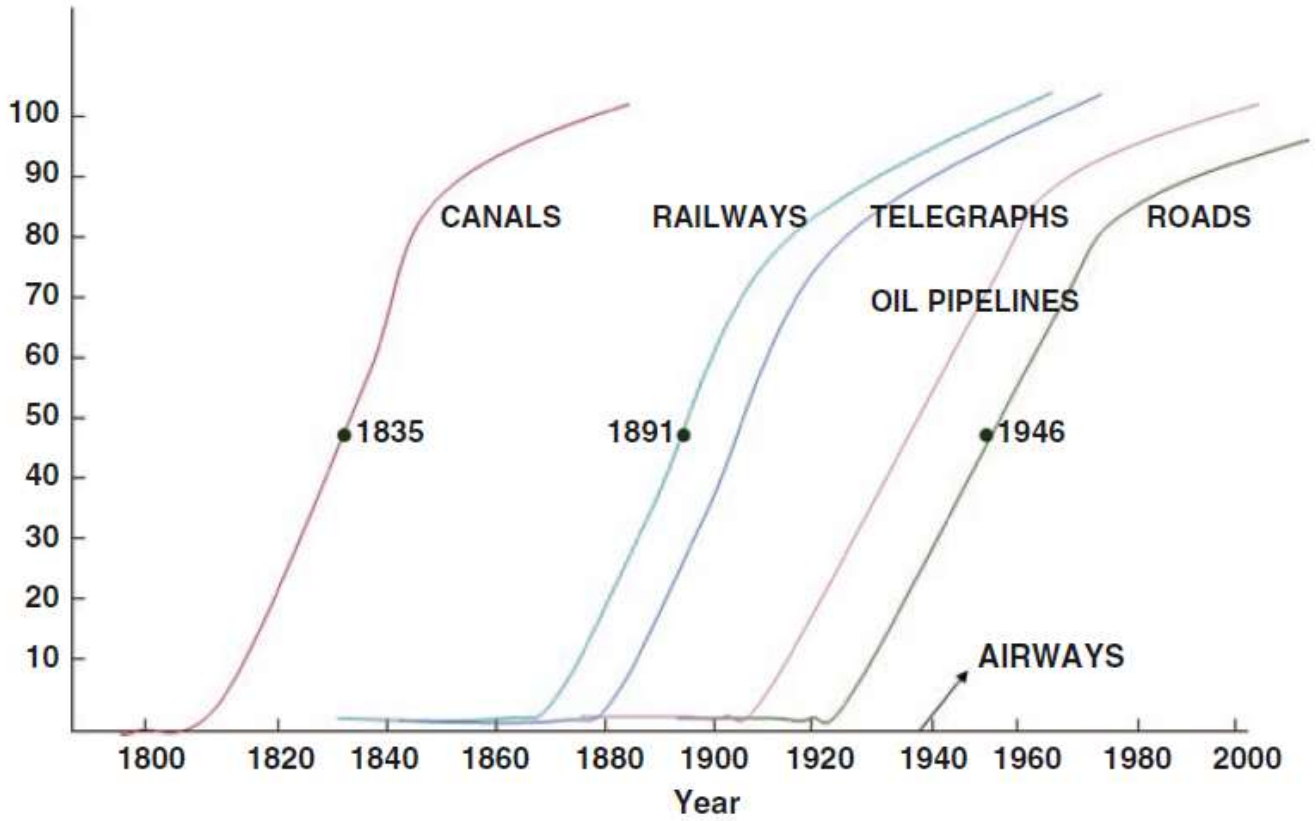


Fig. 6.8 Innovation diffusion patterns in United States. (Source: Grubler 2000)

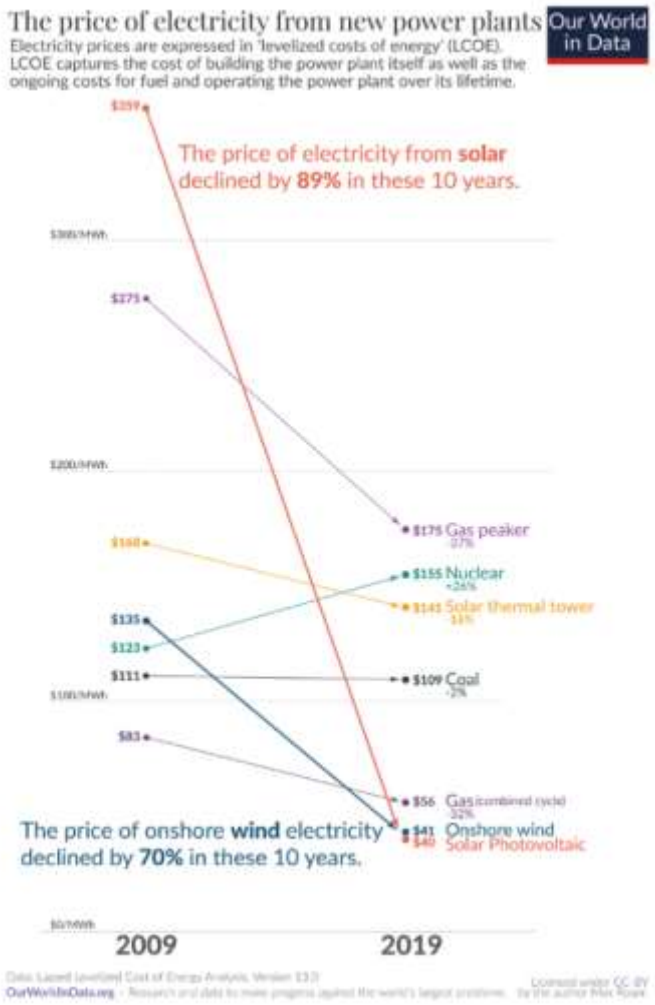
Same patterns seen for

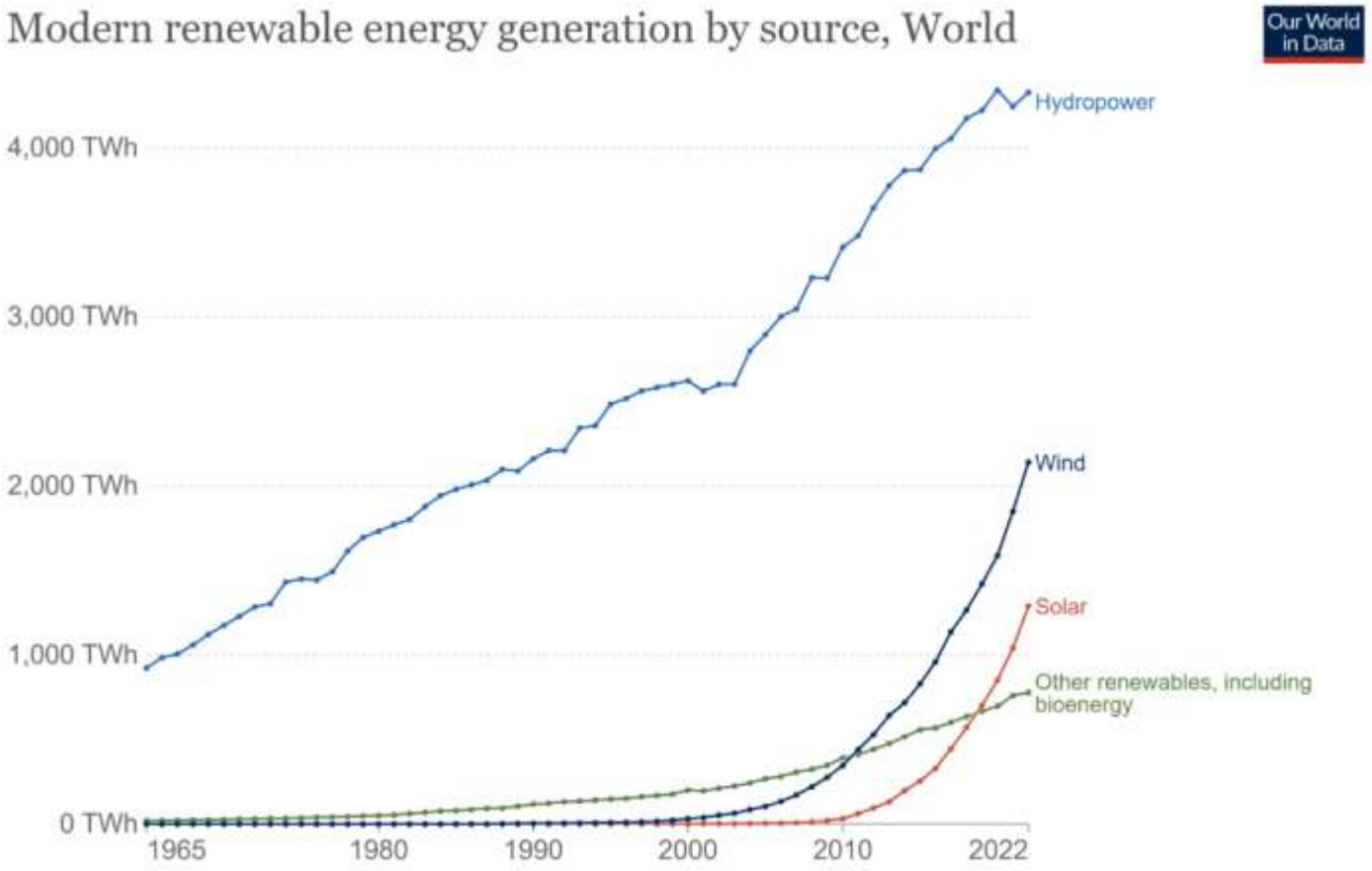
- Telephones
- Computers
- Internet
- Mobile phones
- Broadband
- Smart phones

Exercise: renewable energy

1. Examine the diffusion of various renewable energy technologies and explore the key factors—social, economic, technical, and cultural—that are likely to influence the speed and success of their diffusion
   - Technologies, actors and economic factors

Technology S-curves





Source: Ember's Yearly Electricity Data; Ember's European Electricity Review; Energy Institute Statistical Review of World Energy
OurWorldInData.org/renewable-energy • CC BY

Bluetooth

- Adoption , Diffusion, Licensing

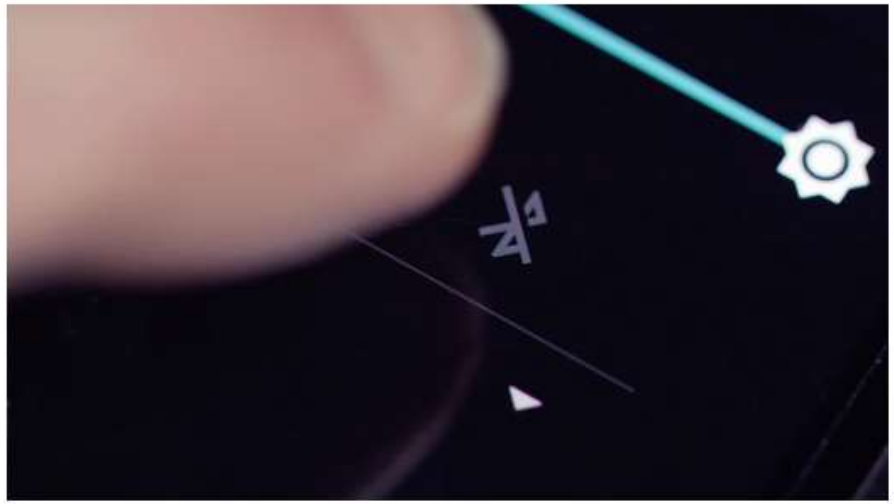
1998

Bluetooth – a new dimension in wireless connectivity

One of Ericsson's best-known innovations, this wireless technology standard for exchanging data over short distances has become ubiquitous since the first specification was released in 1999. In 2013, more than 2.5 billion products containing Bluetooth technology were shipped.

Named after the Danish king Harald "Blåtand" (Bluetooth) Gormsson, the technology was originally intended to allow mobile phones to connect with each other without using cables.

It was quickly discovered that the chip-and-antenna pairing could communicate with a lot more than mobile phones, and today Bluetooth is used in everything from portable speakers to cars, pairing devices into powerful combinations.



Bluetooth

- Launched in 1999 by Ericsson and eight other prominent companies that formed and supported the original Bluetooth Special Interest Group (including Intel, Nokia, Toshiba, and IBM), Bluetooth technology has since become freely accessible to all 7,000 corporate members of the Bluetooth Special Interest Group.
- The first mobile phone equipped with Bluetooth capabilities emerged in 2001, entering a competitive landscape alongside other technologies like 802.11b (WiFi).
- Approximately 175 Ericsson engineers in Lund, Sweden, dedicate their efforts full-time to Bluetooth development, emphasizing the challenge of integrating Bluetooth and other wireless technologies in devices such as laptops, requiring protective measures to prevent signal interference.
- Cahner's In-Stat Group in Newton, Massachusetts, forecasts a promising future for Bluetooth, estimating that by 2005, around 650,000 mobile phones and 300,000 handheld devices will feature Bluetooth technology.
- According to the Centre for Business History in Stockholm, Bluetooth technology enjoys protection from approximately 10 patents, with the majority owned by Ericsson. Notably, U.S. Patent No. 6590928, titled "Frequency Hopping Piconets in an Uncoordinated Wireless Multi-User System," has brought acclaim to Jaap Haartsen, one of the key figures behind Bluetooth's development.

IPR

- World Intellectual Property Organisation (WIPO), intellectual property (IP) refers to 'creations of the mind, such as inventions; literary and artistic works; designs; and symbols, names and images used in commerce'
- IP rights must be formally recognised through a process of registration and protection.
  - some IP rights arise automatically as soon as a piece of work is created so long as there is a record of this creation process
- Intellectual property is divided into two categories:
  - Industrial IPR includes patents for inventions, trademarks, industrial designs and geographical indications.
  - Copyright covers literary works (such as novels, poems and plays), films, music, artistic works (e.g., drawings, paintings, photographs and sculptures) and architectural design

Why care about mangement of intellectual property?

- Intellectual property rights (IPR) affect the processes of diffusion and adoption
- **Protection** of IPR is central in discussions of collaboration and open innovation
- **Ownership** of IP – or the rights associated with IP – allows individuals or companies to enjoy financial benefits from the sale or licensing of this IP.
- IP can be formally listed within the **balance sheet** of a business and requires a business to develop a formal strategy for its protection
- IPR is important to show **investors** and **collaborators**

BUSINESS SCHOOL
AALBORG UNIVERSITY

Types of IP

- **Patents**: For new or improved products or processes capable of commercial or industrial application, requiring **formal** registration;
- **Trademarks**: For words, symbols, pictures, sounds, smells or a combination of these, to distinguish the goods and services of one trader from those of another, and requiring **formal** registration;
- **Designs**: For the shape or appearance of manufactured goods of the whole or a part of a product resulting from the features of, in particular, the lines, contours, colours, shape, texture or materials of the product itself or its ornamentation, and requiring **formal** registration;
- **Copyright**: For original material in literary, artistic, dramatic or musical works, films, broadcasts, multimedia and computer programs, and requiring **no formal** registration;
- **Circuit layout rights**: For the three-dimensional configuration of electronic circuits in integrated circuit products or layout designs, and requiring **no formal** registration;
- **Plant breeder's rights**: For new plant varieties, and requiring **formal** registration;
- **Confidentiality/trade secrets**: Including know-how and other confidential or proprietary information and requiring **no formal** registration process.

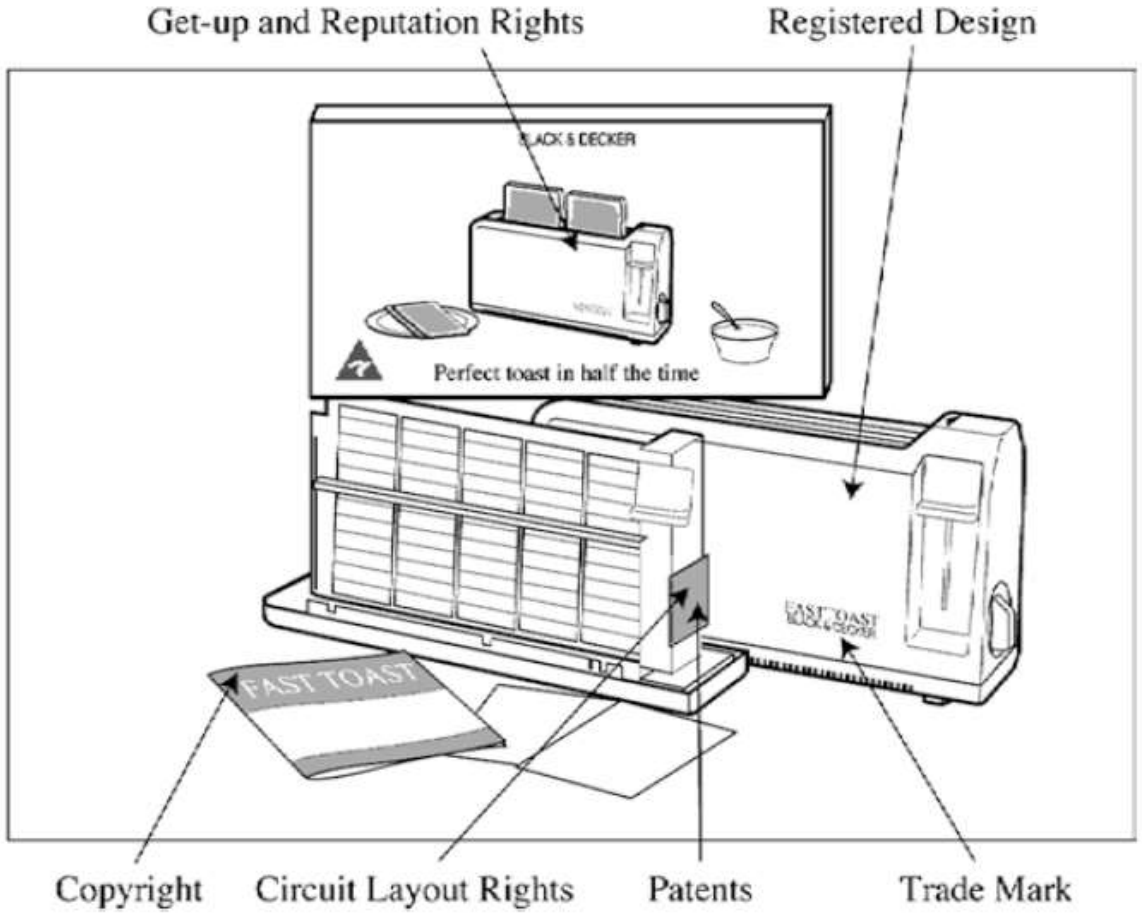


Fig. 13.1 What are intellectual property rights? (The Black & Decker Fast Toast)

Types of knowledge in a firm

- **Tacit** knowledge versus **codified** knowledge
- intellectual **capital**: all forms of knowledge
- intellectual **assets**: knowledge as can be identified as the exclusive right to the creators. This includes material (explicit knowledge) such as software, databases, algorithms, images, documents or manuscripts
- intellectual **property**: that comprises patents, trademarks, copyright and trade secrets

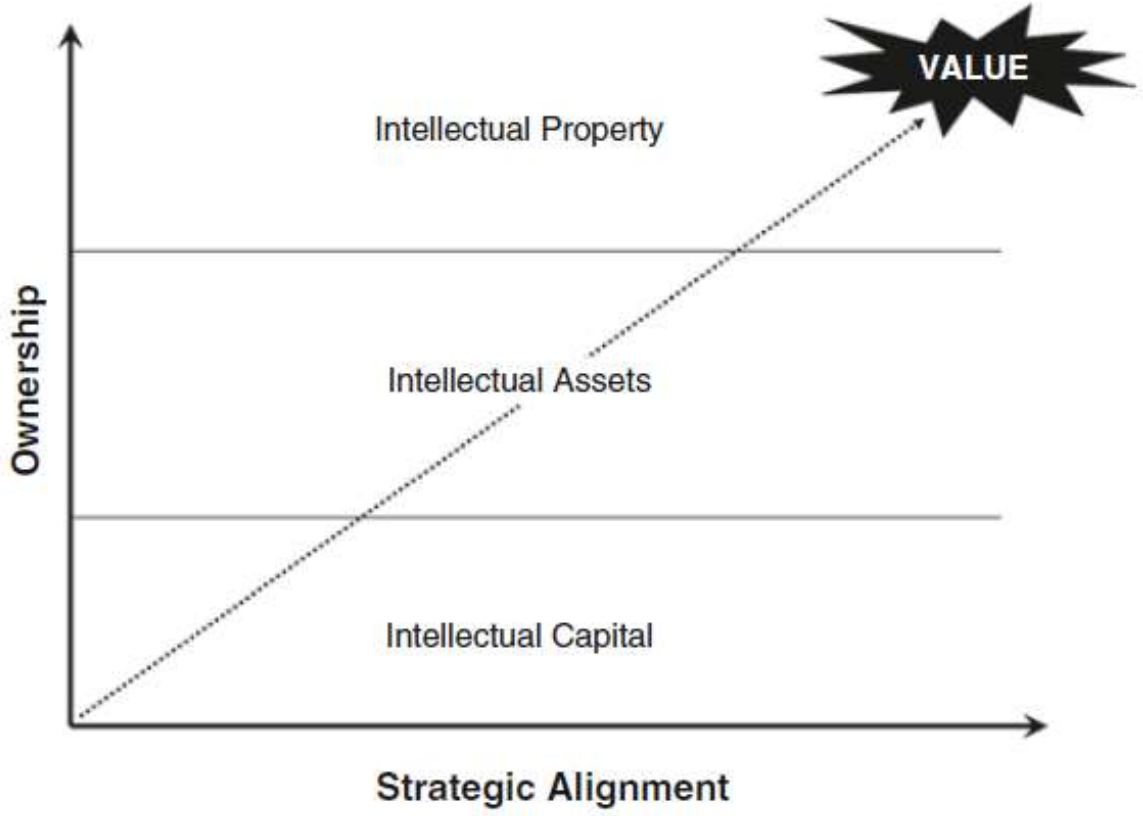


Fig. 13.2 IC, IA, IP degrees of ownership and strategic alignment. (Source: W Bukowitz 2001)

The innovation process and IPR

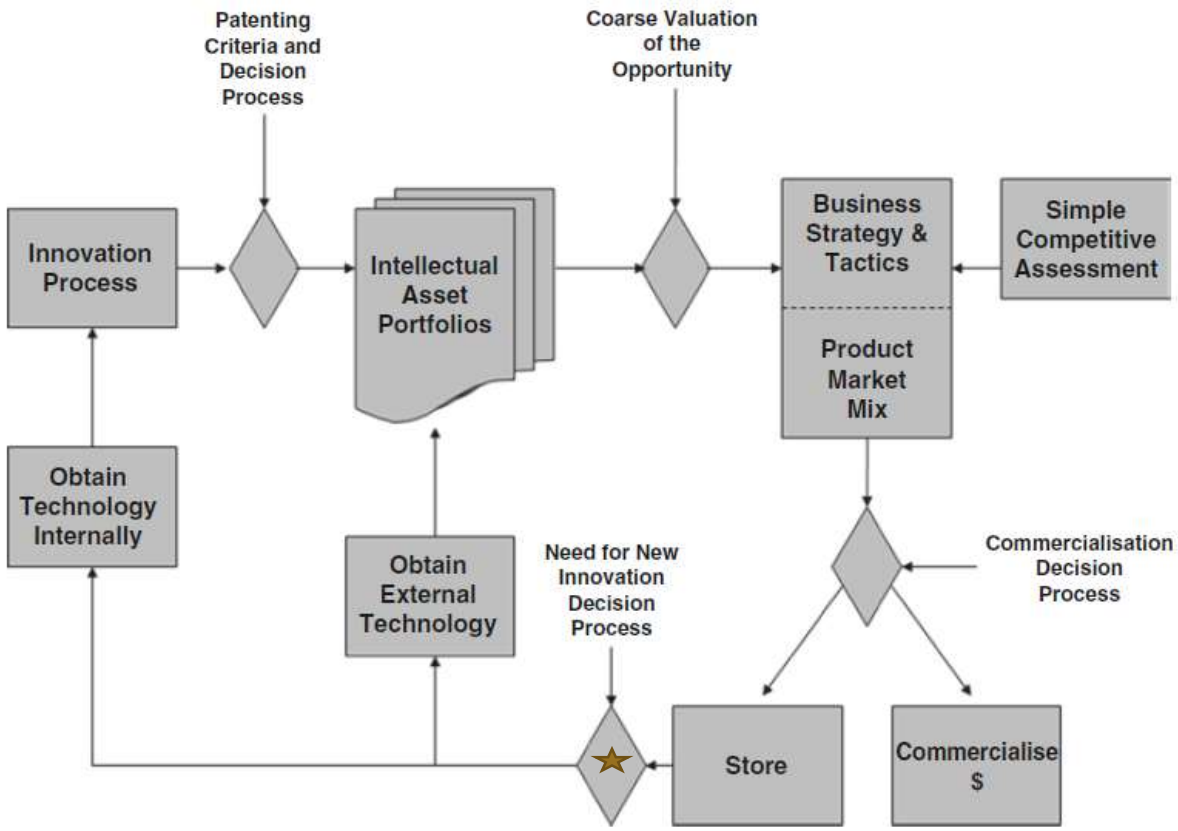


Fig. 13.3 Intellectual asset management system. (Source: Harrison and Sullivan 2000)

Formal IP rights

- Formal IP rights are conferred by government authorities in each country/countries. These rights require the rights holder to formally register their IP with the appropriate authority (e.g., European Patent Office) and pay a fee.
- Patents are expensive
  - cost of registering a standard patent + patent attorney fees + registration costs. Over the 20-year life of the patent, there are likely to be additional fees.
  - patents are only a protection if they can be defended legally.
- Trademarks are less expensive

Patents

- A patent is, ... an exclusive right granted for an invention – a product or process that provides a new way of doing something, or that offers a new technical solution to a problem. A patent provides patent owners with protection for their inventions. Protection is granted for a limited period, generally 20 years. (WIPO 2017 p. 5)
- Purpose: This **official right** is granted to the **patent owner** by the sovereign state for exclusive title to the **use, sale and licencing** of a device, substance, method or process. This legally enforceable right provides the patent owner with the **exclusive right** to commercially exploit the invention for the **life of the patent** once it has been registered and granted to them.
- The documentation generally must have a **clear description** of the invention, including drawings that **provide sufficient detail for a skilled person** in the area of technology to **perform the invention themselves**.
- It should also make claims to define the scope of protection that is being sought. These details are then used to determine future claims made for or against the legal protection offered by the patent.

Patents

1. It must be new.
   - The process or invention must demonstrate that it is genuinely new and does not form part of the 'state of the art', or what was already available to the public prior to the patent application. This means that if the idea has been published, presented at conferences, illustrated in a public display or generally disseminated prior to the patent application, it would not qualify;
2. It must involve an inventive step.
   - It should not be 'obvious' from the perspective of a person skilled or knowledgeable in the discipline or area of technology from which the invention is derived;
3. It must have industrial application.
   - The invention or process should be able to be made use of in industry in some demonstrable way.

- Disclosure of the process in the public domain is likely to void the right to a patent.
- Make use of written confidentiality agreements in discussions with business partners and advisors prior to the filing of a patent to protect patentable IP

2022: Lars has invented the coffee lid of the future - is already worth 65 million

- He has invented a coffee lid made entirely of wood fibers. It feels like plastic, but for confusion it is sustainable and contains no microplastic.

- He has received a patent for the lid in several countries, and the latest assessment of the company, which exclusively deals with the lid, is DKK 65 million.
- Lars Bendix has sent the new lid out to potential buyers all over the world, and he himself says that the feedback has been really positive.
- Some will buy license, some will buy the lid, the influx is just really big. Now we have to have a lot of money in and build up a production, says Lars Bendix, Danish Coffee.

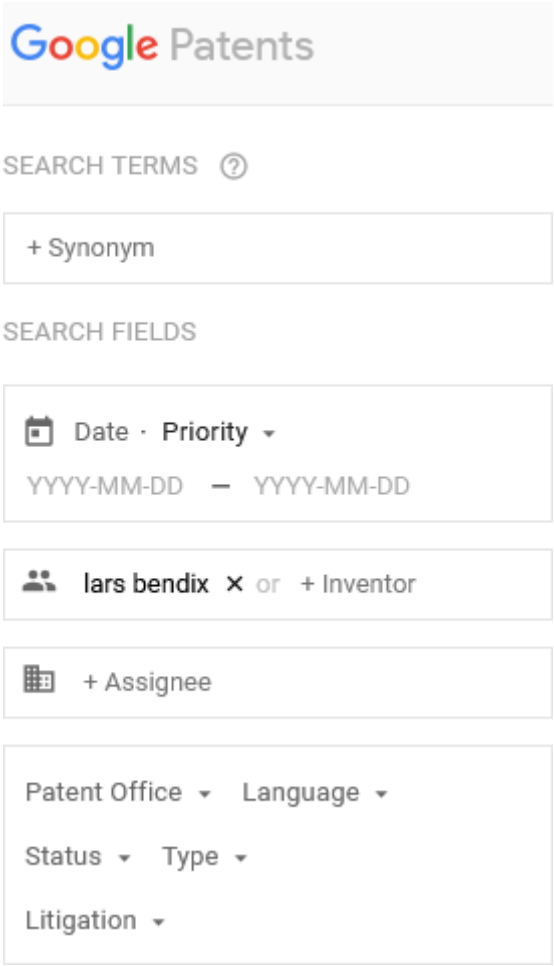
2024: the coffee lid

- The 64-year-old inventor and entrepreneur Lars Bendix is finally seeing his much-talked-about coffee lid, Liplid, launched. B Company from Odense has secured exclusive distribution rights in Denmark, and there is already high demand.
- Liplid, made from wood fibers, is designed to prevent lip burns and is completely plastic-free. It uses 24% less material compared to similar lids. The idea for Liplid came during a lunch meeting between Bendix and his collaborator, Håkan Löfholm, in Stockholm in 2016.
- Here, a quick sketch was made for a new coffee lid to address the challenges posed by existing disposable lids on the market. While Håkan Löfholm secured patents and initiated discussions with potential investors, Lars Bendix went back to the drawing board to further design the to-go lid.
- Investor Tahir Siddique, known from "Løvens Hule," joined the project, and B Company pre-sold over 300,000 units before Liplid even arrived at their warehouse. The lid is now available in Denmark and Norway, with plans to expand to other markets.
- Bendix expressed satisfaction with the launch, highlighting its contribution to reducing plastic use and the strong demand even before its official release.

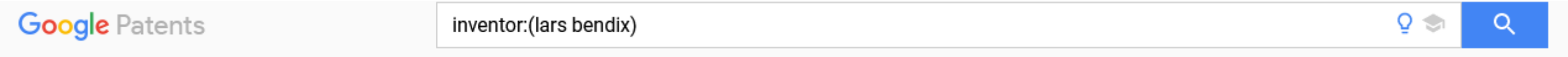
Exercise

13.5.2 Checklist for Protecting IP Assets

10. Search the patent, trademark and design databases as well as other literature and the internet to ensure that your ideas are new, and to avoid infringing the rights of others. You can also search for new business opportunities as well as keep a tab on what your competitors are doing.



- **Find Lars Bendix's patent**
- https://patents.google.com/



Exercise

1. Why is it beneficial for Lars Bendix to obtain a patent for his coffee lid invention?
2. What are the advantages of securing a patent for Lars Bendix and his company?

- Discussion on challenges to Intellectual Property Rights (IPR) for the Coffee Lid Patent:

3. Are there any hurdles in effectively protecting the intellectual property rights (IPR) associated with the coffee lid patent?

Trademarks

- A **trademark**, … is a **distinctive sign** that identifies certain goods or services produced or provided by an individual or a company. Its origin dates back to ancient times when craftsmen reproduced their signatures, or marks, on their artistic works or products of a functional or practical nature.
- Over the years, these marks have evolved into today's system of trademark registration and protection.
- The system helps consumers to identify and purchase a product or service based on whether its specific characteristics and quality – as indicated by its unique trademark – meet their needs (WIPO (2017) p 8).
- It can be a letter, number, word, phrase, sound, smell, shape, logo, picture, aspect of packaging or any combination of these.

Designs

- An industrial design is “the ornamental or aesthetic aspects of an article. A design may consist of three-dimensional features, such as the shape or surface of an article, or two-dimensional features, such as patterns, lines or colour” (WIPO (2017), p.12).
- It often refers to the visual aspects of a logo or product resulting from, in particular, the features of the lines, contours, colours, shape, texture or materials of the product or its ornamentation.
- The registration of a design grants the owner the exclusive rights to make use of a product design, and to licence the use of this design or to sell it.
- patent relates to the function of a device or product, a registered design relates to its form or appearance

Copyright

- Copyright
  - Copyright gives the creators of a wide range of material – such as literature, art, music, sound recordings, films and broadcasts – economic rights enabling them to control the use of their material in a number of ways, such as by making copies, issuing copies to the public, performing in public, broadcasting and using the material online.

Confidentiality and Trade Secrets

- Confidentiality agreement
- Non disclosure
  - The usual approach is for the company to require all employees and third-party actors (e.g. suppliers, sub-contractors) to sign confidentiality or non-disclosure agreements, thereby restricting their right to reveal trade secrets or proprietary knowledge during their employment or association with the business.
- Non compete
  - contractors and employees have written undertakings not to compete with your business after they leave – in addition to signing a confidentiality agreement

A Danish company wants to file for a patent on a new chemical process for manufacturing plastics. According to the course material, which of the following conditions must be met for the patent to be granted, and what is a key implication of receiving it?

Choose one or more answers

- [ ] Upon granting the patent, the company receives a temporary monopoly on the use of the process in exchange for the full public disclosure of how the process works.
- [ ] The process must be expected to be a commercial success and profitable within five to ten years or else the patent cannot be extended.
- [ ] The company must prove it has plans to build or outsource to a factory capable of implementing the process on an industrial scale.
- [ ] Once granted, the patent provides permanent, worldwide protection that is enforced by patent offices.

Control over resources

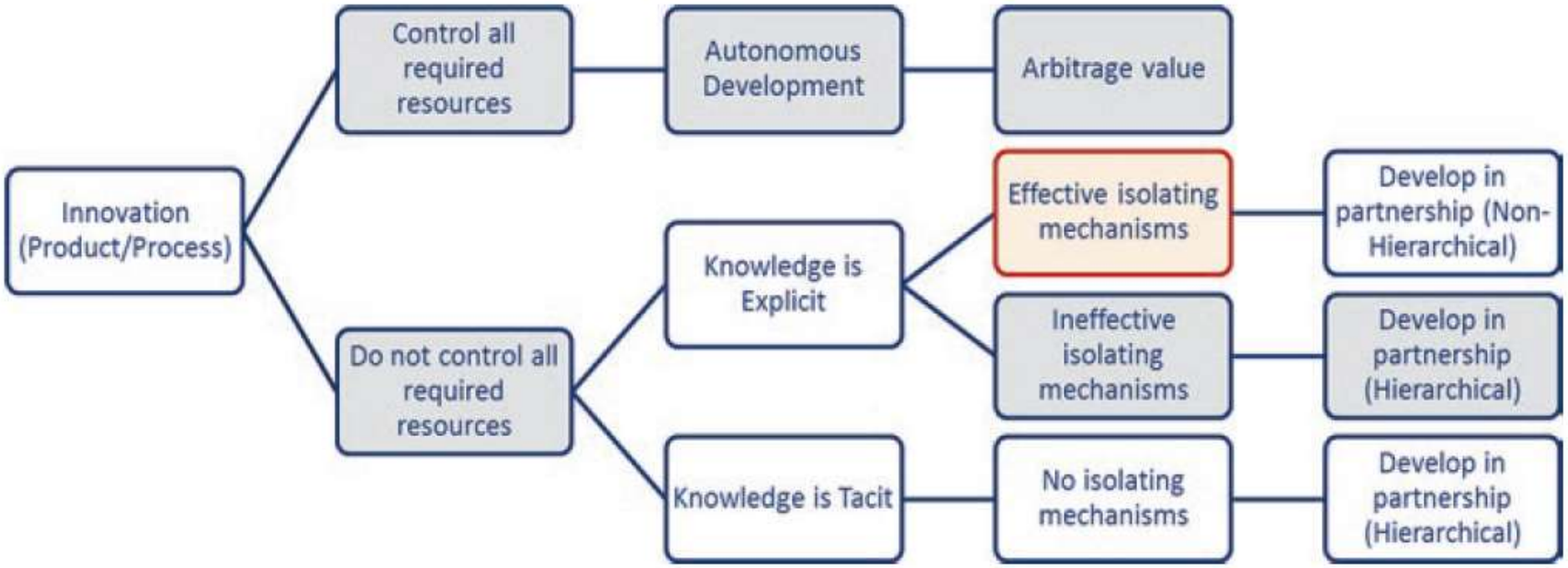


Fig. 13.4 Resources as isolating mechanisms in commercialisation. (Source: Alvarez and Barney 2004)

Other types of protection

- Causal ambiguity
  - **Tacitness**: This is the implicit and non-codifiable accumulation of skills and knowledge that is generated from learning by doing;
  - **Complexity**: This is the possession of a large number of interdependent skills and assets (resource bundles);
  - **Specificity**: This is the transaction-specific skills and assets that are utilized in the production processes and service delivery of products or services.

Licencing

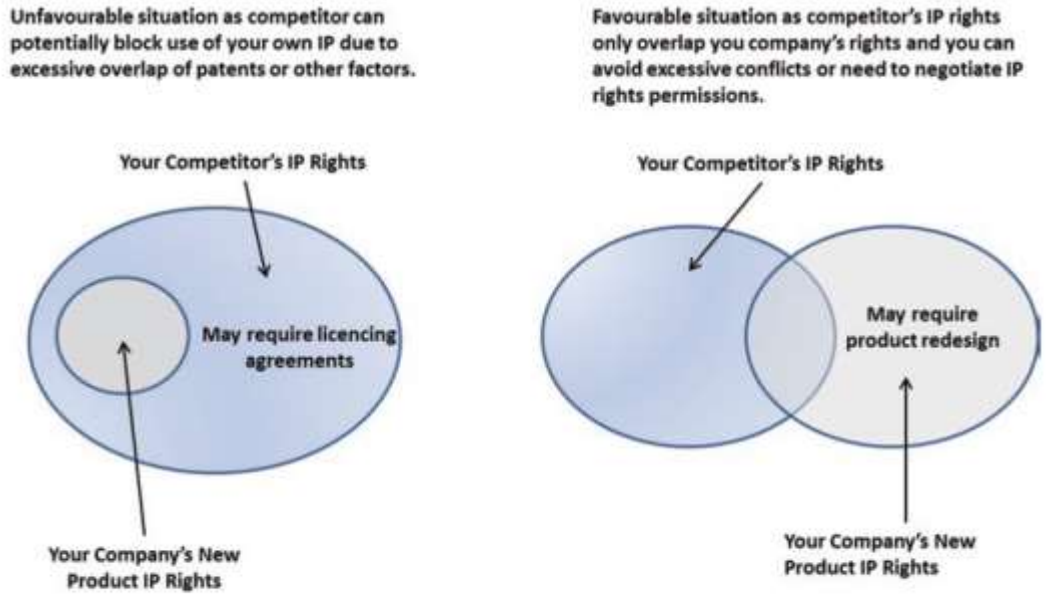


Fig. 13.5 The boundaries of IP rights. (Source: Ambrozy 2013)

Table 13.1 Factors favouring licencing parties

Factors favouring the licensor	Factors favouring the licensee
Broad, valid, assignable patents	Substantial NPD and product commercialisation work required
Related know-how and trade secrets	Substantial investment in plant and personnel required
Marketing contacts and potential customers	Existing strong competition in both market and product segment
Existing brand names	Existing plant and capacity
Company reputation in the market	Existing skilled workforce
Well-funded and resourced	Existing sales and distribution channels
Productive R&D pipeline for NPD.	Substantial NPD and product commercialisation work required
History of successful licensing	Substantial investment in plant and personnel required

Source: QPSX (2005)

Valuation

Considerations in IP Assets Valuation

- What is the lifecycle of the technology in which the IP asset resides?
- Is their legal protection of the IP asset (e.g. patents)?
- What is the economic life of the IP asset?
- How transferable is the IP asset from one owner to the next?
- Are there any restrictions on the exploitation of the IP asset?
- What is the level of R&D required to maintain the IP asset?
- What is the nature of the competitive environment?
- What is the normal financial return associated with the development of such an IP asset?
- What is the extent of functional and/or technological obsolescence?
- What are the forecasts for economic and technological trends in the industry?
- What is the likely cost of developing competing IP assets?

Source: Bertolotti (1995).

Innovation and Competition in the Plant-Based Milk Market

- The market for plant-based milk has grown rapidly, attracting both specialist firms and traditional dairy producers. Several companies now compete within this emerging category—each with different capabilities, timing of entry, and strategic positions

- Several Key Players
- First Movers – Specialist Non-Dairy Firms
  - Oatly
    - Origin: Sweden
    - Core competency: Plant-based, non-dairy food technology
    - Known for: Oat-based milk innovations, strong brand identity, patented enzyme technology
    - Strategy: Differentiation through sustainability branding + technology leadership
  - Naturli'
    - Origin: Denmark
    - Established: 1988 as a plant-based specialist
    - Ownership: Acquired in 2010 by Orkla, a large Norwegian company
    - Competency: Long non-dairy heritage + later scaling through Orkla's distribution
    - Strategy: Broad plant-based portfolio (milk, butter alternatives, meat-free products)
- 2. Followers - Dairy Firms Entering late
  - Arla (Brand: JÖRD)
    - Type: Large multinational dairy cooperative Entered plant-based: 2020
    - Competency: Strong distribution channels, production scale, retail presence
    - Strategy: Fast imitation + leveraging trust and shelf access
  - Thise
    - Type: Danish organic dairy cooperative Entered plant-based: 2020
    - Competency: Strong organic brand reputation, but limited non-dairy experience
    - Strategy: Niche positioning
  - Dryk
    - Type: Small independent plant-based firm
    - Competency: Agility + focus on "premium barista" products
    - Strategy: Competing through quality and specialty formats

Group discussion questions

- Who will win?
  - Discuss using concepts from diffusion theory, IPR protection, and market structure.
- Discuss in groups and explain who is likely to benefit from the innovation
  - Search for additional information online
  - Who captures the most value?
  - Consider first-mover advantages vs. follower scale advantages.
  - Which firms benefit most from innovation?
    - Look at:IPR; Branding; Diffusion through retail channels; Market power
- Is the future dominated by specialists or traditional dairy firms? Why?

- Scenario: Seven years ago, a small start-up company opened VegaNiCE, a shop selling vegan ice cream in a large city in Denmark. The company's products have become popular, and it has opened a couple of extra shops. This has resulted in a growth in turnover, but not in profits. The company produces the vegan ice cream (based on coconut milk) with various flavors in small batches in the shops. A particular popular product is a vegan ice cone with a vegan chocolate covered marshmallow.
- The market for vegan products in Denmark is growing rapidly, which has opened some interesting opportunities for the company. The company has decided to move into the retail market with their vegan ice cream. However, selling their products in supermarkets requires a change in their production process. Therefore, they have acquired an old dairy factory and new ice machines that allow for a production of 130 liters of vegan ice per hour. The new production facilities also include storage and allow the company to ramp up their production. The company is also experimenting with new flavors of ice cream and has now developed more than 30 different flavors. However, a growing market for vegan products also attracts new competitors and pose new challenges for the company.
- 1) Based on the information above, how would you characterize the innovation type(s) of VegaNICE?
- 2) In general, are there particular innovation activities for which small firms are likely to outperform large firms, and particular innovation activities where large firms can outperform small firms?
- 3) Identify the main factors that affect VegaNICE's possibilities to protect and capture rents of their innovations when entering the retail market.

What have you learned today

- Diffusion of innovation
  - E.g. S-curves
- Adoption of innovation
  - TAM and TAM2 models
- What is IPR , how is it used and managed